

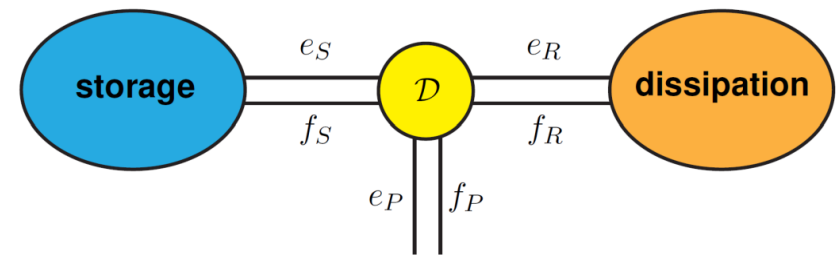
Port Hamiltonian Systems

PHS linked via pairs (f,e) of equal dimension vectors of flow and effort variables.

- Energy-storing multi port element: S (storage)
- Energy-dissipating elements: R (resistive)
- Energy-routing elements. D (Dirac)

Three ports

- Energy storage port (f_S, e_S)
- Energy-dissipation port (f_R, e_R) , and
- External port (f_P, e_P) , for external interaction of system with its environment.



Products denote energy transmission: $e_S^T f_S, e_R^T f_R, e_P^T f_P$

Port Hamiltonian Systems

$$\begin{aligned}\dot{x} &= [J(x) - R(x)]\nabla_x H(x) + G(x)u \\ y &= G(x)^T \nabla_x H(x)\end{aligned}$$

- $x \in R^n$ is n dimensional vector representing the system's state
- $H(x)$ is the system's Hamiltonian function
- $J(x) \in R^{n \times n}$ is the skew symmetric interconnection matrix
- $R(x) \in R^{n \times n}$ is the symmetric positive semi definite dissipation matrix
- $G(x) \in R^{n \times m}$ is control input matrix
- $u \in R^m$ is m dimensional control input vector and $y \in R^m$ is output vector

$H(x)$: system's energy, $J(x)$: energy-conserving interactions between the various components of the system and $R(x)$ how these components dissipate energy.

1. Incrementally port-Hamiltonian systems

Port Hamiltonian systems are composed of Dirac structures.

The basic idea is replace the composition of a Dirac structure D and an energy-dissipating relation R by a *maximal monotone relation* M .

Dirac structure $D \subset \mathcal{F}_S \times \mathcal{E}_S \times \mathcal{F}_P \times \mathcal{E}_P \times \mathcal{F}_R \times \mathcal{E}_R$
with Hamiltonian $H: X \rightarrow R$

General Port Hamiltonian Systems

$$\dot{H} = -e_S^T f_S = -\dot{x} \nabla H(x), f_S = -\dot{x}, e_S = \nabla H(x) \text{ for state } x \in X$$

Power balance with external port is given by

$$e_S^T f_S + e_R^T f_R + e_P^T f_P = 0$$

Energy dissipation equation with resistive relation $e_R^T f_R \leq 0$ is given by

$$\dot{H} = e_R^T f_R + e_P^T f_P \leq e_P^T f_P$$

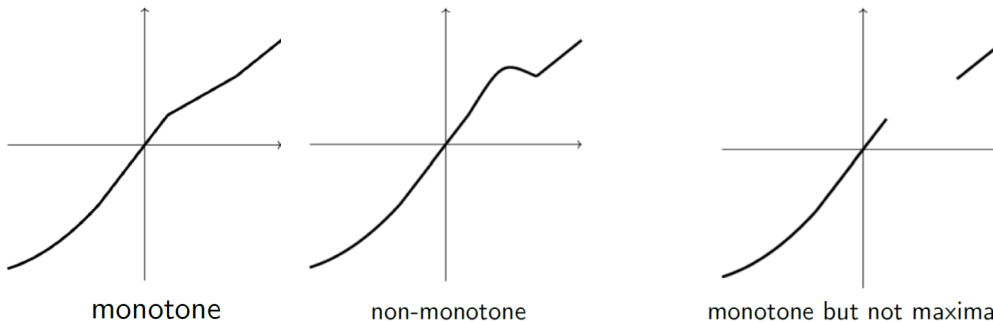
This inequality expresses that the increase of the internally stored energy is always less than or equal to the externally supplied power.

Monotone Relation

- A relation $\mathcal{M} \subseteq \mathcal{F} \times \mathcal{E}$ is said to be *monotone* if
$$\langle v_1 - v_2 | u_1 - u_2 \rangle \geq 0$$

For all $(u_i, v_i) \in \mathcal{M}$ with $i \in \{1, 2\}$

It is called maximal monotone if it is monotone and $\mathcal{M}'$ is monotone (for vector valued function) and $\mathcal{M} \subseteq \mathcal{M}' \Rightarrow \mathcal{M} = \mathcal{M}'$ holds.



Conservative Port Hamiltonian Systems

Consider a maximal monotone relation $\mathcal{M}$ defined on space of

$$\mathcal{M} \subset \mathcal{F}_S \times \mathcal{E}_S \times \mathcal{F}_P \times \mathcal{E}_P$$

flow and effort variables for conservative PHS and a Hamiltonian $H: \mathcal{F}_S \rightarrow \mathbb{R}$.

$\mathcal{F}_S \times \mathcal{E}_S$: (energy –storing elements) and $\mathcal{F}_P \times \mathcal{E}_P$: (external ports)

Dynamics of the corresponding *incrementally port-Hamiltonian system* is given below for all time constants t .

$$\left(-x'(t), f_P(t), \frac{\partial H}{\partial x}(x(t)), e_P(t) \right) \in \mathcal{M}$$

Conservative Port Hamiltonian Systems

- Dynamics of incrementally port-Hamiltonian systems satisfies incremental '*dissipation inequality*' property.

$$\langle \nabla H(x_1(t)) - \nabla H(x_2(t)) | x_1(t) - x_2(t) \rangle \leq \langle e_P^1(t) - e_P^2(t) | f_P^1(t) - f_P^2(t) \rangle$$

for all quadruples $(x_i, \dot{x}_i, f_P^i, e_P^i), i = 1, 2$ satisfying $(-x'(t), f_P(t), \frac{\partial H}{\partial x}(x(t)), e_P(t)) \in \mathcal{M}$

Every conservative port-Hamiltonian system is also incrementally port-Hamiltonian as follows from the following **lemma**.

Lemma

Lemma: Every Dirac structure on $\mathcal{F} \times \mathcal{E}$ is maximal monotone.

Proof.

Let $\mathcal{D} \subset \mathcal{F} \times \mathcal{E}$ be a Dirac structure.
Let $(f_i, e_i) \in \mathcal{D}$ with $i = 1, 2$.

Since $\langle e | f \rangle \geq 0$ for all $(f, e) \in \mathcal{D}$,
we get

$$\langle e_1 - e_2 | f_1 - f_2 \rangle = 0.$$

Therefore, $\mathcal{D}$ is monotone on $\mathcal{F} \times \mathcal{E}$. Let $\mathcal{D}'$ be a monotone relation on $\mathcal{F} \times \mathcal{E}$ such that $\mathcal{D} \subseteq \mathcal{D}'$. Let $(f', e') \in \mathcal{D}'$ and $(f, e) \in \mathcal{D}$. Since $\mathcal{D}'$ is monotone, $\mathcal{D} \subseteq \mathcal{D}'$, and since $\mathcal{D}$ is a subspace, we have $0 \leq \langle e' - \alpha e | f' - \alpha f \rangle = \langle e' | f' \rangle - \alpha(\langle e' | f \rangle + \langle e | f' \rangle)$ for any $\alpha \in \mathbb{R}$. This means that

$$\langle e' | f \rangle + \langle e | f' \rangle = 0$$

and hence $(f', e') \in \mathcal{D}^{\perp\perp} = \mathcal{D}$. Therefore, we get $\mathcal{D}' \subseteq \mathcal{D}$. Since the reverse inclusion already holds, we further get $\mathcal{D}' = \mathcal{D}$. Consequently, $\mathcal{D}$ is maximal monotone. ■

Relation with Convex Function

Consider a Hamiltonian $H: X \rightarrow \mathbb{R}$, and a convex function $K: X^* \times U \rightarrow \mathbb{R}$, then the system

$$\dot{x} = -\frac{\partial K}{\partial e}(\nabla H(x), u), \quad e = \nabla H(x) := \frac{\partial H}{\partial x}(x)$$
$$y = \frac{\partial K}{\partial u}(\nabla H(x), u)$$

is a maximal cyclically monotone port-Hamiltonian system. Special case occurs if the convex function $K(e, u)$ is of the form

$$K(e, u) = P(e) + e^T B u$$

with P a convex function of e , and B an $n \times m$ matrix gives another class of system.

$$\dot{x} = -\frac{\partial P}{\partial e}(\nabla H(x)) - B u, \quad y = B^T \nabla H(x)$$

Gradient algorithm in continuous time

Consider the problem of minimizing a convex function $P: \mathbb{R}^n \rightarrow \mathbb{R}$.

Gradient algorithm in continuous time is given as

$$\tau \frac{dq}{dt} = -\frac{\partial P}{\partial q}(q) - Bu$$
$$y = B^T q$$

τ is a positive definite matrix determines the time-scales of algorithm.
 q : state, B : $n \times m$ matrix, $u \in \mathbb{R}^m$: input vector represent interaction with other algorithms defining a conjugate output vector as $y = B^T q \in \mathbb{R}^n$.

Extended class of maximal monotone PHS

$$\begin{aligned}\dot{x} &= J\nabla H(x) - \frac{\partial P}{\partial e}(\nabla H(x)) - Bu \\ y &= B^T \nabla H(x)\end{aligned}$$

where J is a skew-symmetric matrix, and P a convex function as above.

State vector $x := \tau q$, B : $n \times m$ matrix, quadratic Hamiltonian,

$$e_S = \nabla H(x) = \tau^{-1}x \text{ and } H(x) = \frac{1}{2}x^T \tau^{-1}x.$$

If $J \neq 0$ then the underlying maximal monotone relation is not derivable from a convex function, and the system is not cyclically monotone port-Hamiltonian anymore.

Relevance to possibly use this systems using incremental PHS framework.

2. PORT–HAMILTONIAN GRADIENT FLOW

(PHGF): A novel optimization framework inspired by the port Hamiltonian framework equipped with implicit convergence guarantees.

The proposed formalism can be leveraged to either inspire new optimization approaches equipped with implicit convergence guarantees, or to offer improvements to existing solutions.

The design of ad–hoc inertia matrices M allow for the introduction of **interconnections** between parameters via **coupling terms**, which can affect convergence dynamics.

PHGF

- A general framework for continuous–time gradient descent, often referred to as gradient flow.
- Extend Hamiltonian gradient flows to neural network parameters.
- In **continuous–time**, PHGFs allow adaptive ODE solvers to stick to the energy manifold, avoiding divergence due to large learning rates.
- In **fixed–step** optimization, on the other hand, PGHFs open the door to novel fixed–step approaches based on symplectic discretization's.

Symplectic integrator

- They are designed for the numerical solution of Hamilton's equations.
- It preserve energetic properties of the discretized model.

PHGF

- A port–Hamiltonian gradient flow can be represented symbolically by stochastic differential equation (SDE) in the form

$$d_X = F \nabla_X H(X) dt + g(x) dw$$

- $w(t)$ denotes a multivariate Wiener process, F is the system matrix.
- $X := (\theta, w)$
- $g: \mathbb{R}^{2d} \rightarrow \mathbb{R}^{2d}$ approximate function

SYMPLECTIC DISCRETIZED PHGF

- Stormer–Verlet Port–Hamiltonian Gradient Flow (SV–PHGF)
- *SV-PHGF* iterative update rule is given by

$$\omega_{t+1} = 2 - \frac{\gamma}{(2 + \gamma\beta)\omega_t} - \frac{2\gamma}{2 + \gamma\beta} \nabla_{\theta} l_{\theta_t}(x_t)$$

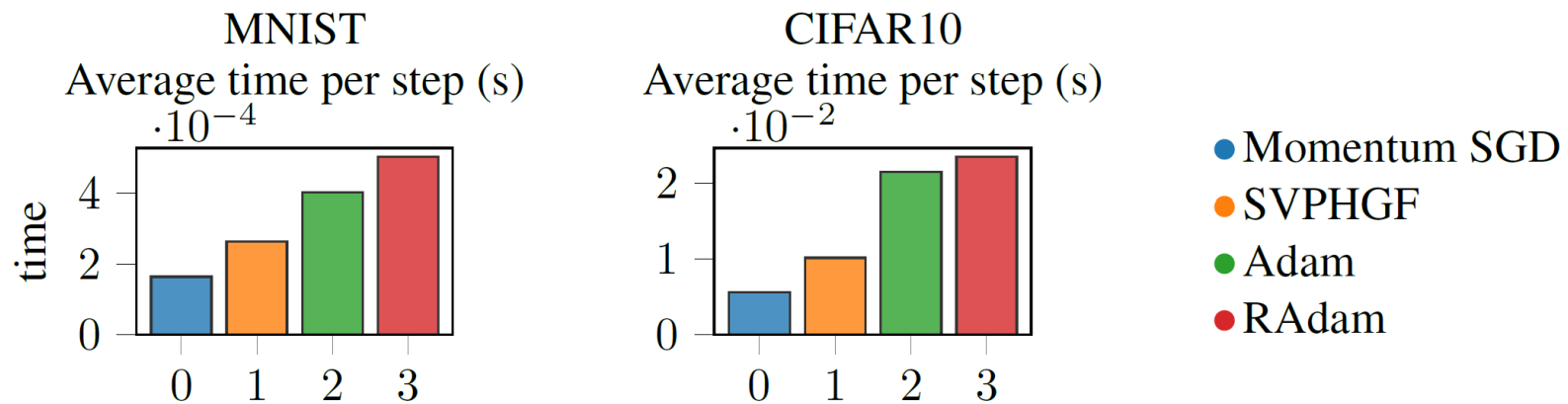
$$\theta_{t+1} = \theta_t + \gamma M^{-1} \omega_{t+1}$$

- SV–PHGF leverages separability by construction of the generalized energy

$$H_t := l_{\theta_t}(x_t) + \frac{1}{2} \omega_t^T M^{-1} \omega_t$$

- To adapt the SV uses implicit second order method.

Comparison of time per optimization step



- All methods are tasked with optimizing the same architecture:
- A 2 layer fully-connected NN with 1 million parameters for MNIST and a ResNet 56 with 1 million parameters for CIFAR10.
- **SV-PHGF** is computationally more efficient than both Adam and RAdam. As MNIST was trained with batch size 128 and CIFAR10 256, the result shows how the difference between optimizers remains significant for larger batches.

Thanks